

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 12: Supplement: Hilbert Variational Gravitation

Munhyeok Kang

2026-11-16–2026-11-22

Overview

The Hilbert action is derived and compared with the course model at the level of assumptions, field equations, and empirical coverage.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$S_H = \int d^4x \sqrt{-g} R$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$G_{\mu\nu} = \kappa T_{\mu\nu}$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$N_{pass} = 3/8$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Hilbert, Variational Gravitation, campus ed., §§1–4
- Historical Model Comparison H-1916
- Mathematics cross-list note